

Journals Published

1. **Mukhopadhyay, S.;** Das, A.; Rajput, S.; Prajapati, D. J.; Nembenna, S., Reactivity of Bis(Guanidinate) Magnesium Hydride and Alkyl Complexes: Mg-E (E = S, Se, Te) Bond Formation and Structural Diversification. *Inorg. Chem.* **2026**, *65*, 20064–20077.
2. Rajput, S.; Padhan, S.; **Mukhopadhyay, S.;** Nembenna, S., Neutral and Cationic Bis(guanidinate) Digallanes and Gallium-Catalyzed Alkene Hydroboration. *Inorg. Chem.* **2026**, [10.1021/acs.inorgchem.6c03265](https://doi.org/10.1021/acs.inorgchem.6c03265)
3. **Mukhopadhyay, S.;** Das, A.; Rajput, S.; Behera, G.; Nembenna, S., Mechanistic Studies of Bis(Guanidinate) Magnesium Hydride as a Precatalyst for the Hydroboration of Organic Carbonates and Carbon Dioxide. *Catal. Sci. Technol.* (**2026**), [10.1039/D6CY00898D](https://doi.org/10.1039/D6CY00898D).
4. Thilakan, N. M.; Rajput, S.; **Mukhopadhyay, S.;** Nembenna, S., Zinc-Mediated C–H Borylation of Terminal Alkynes: Mechanistic Insights. *Asian J. Org. Chem.* **2026**, *15* (3), e00637.
5. Padhan, S.; Rajput, S.; **Mukhopadhyay, S.;** Sabharwal, G.; Nembenna, S. Bis-Guanidinate Aluminum Complexes: Synthesis and Catalytic Application in Three-Component Aminophosphorylation of Aldehydes. *Organometallics* **2026**, *45*, 1384–1394
6. **Mukhopadhyay, S.;** Das, A.; Padhan, S.; Nembenna, S. Bulky bis-guanidinate ligand-supported magnesium hydride: Synthesis, reactivity, and catalytic applications. *Inorg. Chim. Acta* **2026**, *589*, 122924.
7. **Mukhopadhyay, S.;** Padhan, S.; Das, A.; Nembenna, S. Bis-guanidinate magnesium amide-catalyzed hydroboration of nitriles and isocyanides via a substrate-assisted pathway. *Org. Chem. Front.* **2026**, *13*, 350–363.
8. Das, A.; **Mukhopadhyay, S.;** Rajput, S.; Nembenna, S. Magnesium-catalyzed hydroamination of carbodiimides and hydroboration of cyanamides. *Dalton Trans.* **2025**, *54*, 17553–17563.
9. Das, A.; **Mukhopadhyay, S.;** Nembenna, S. Conversion of alkynes to imidazolidin-2-ones: Insight into the role of magnesium amido complexes in the reaction. *Organometallics* **2026**, *45*, 36–47.
10. Das, A.; **Mukhopadhyay, S.;** Rajput, S.; Nembenna, S. Catalytic hydroboration of α,β -unsaturated ketones using β -diketiminato magnesium hydride: Mechanistic insights. *Dalton Trans.* **2026**, *55*, 6056–6066.
11. Rajput, S.; Padhan, S.; Thilakan, N. M.; **Mukhopadhyay, S.;** Nembenna, S. Nickel-catalyzed dehydrogenative Zn–Zn coupling to a Zn(I) dimer and its reactivity. *Chem. Commun.* **2026**, *62*, 534–538.
12. Sahoo, R. K.; **Mukhopadhyay, S.;** Rajput, S.; Jana, A.; Nembenna, S. Bulky bis-guanidinate ligand-based neutral and cationic zinc complexes: Synthesis, characterization, and catalytic applications. *Dalton Trans.* **2025**, *54*, 15554–15560.
13. **Mukhopadhyay, S.;** Rajput, S.; Sahoo, R. K.; Nembenna, S. Magnesium-catalyzed C–C coupling of terminal alkynes with carbodiimides: Isolation of active species and mechanistic

insights. *Eur. J. Org. Chem.* **2024**, e202400702.

14. **Mukhopadhyay, S.**; Sahoo, R. K.; Patro, A. G.; Khuntia, A. P.; Nembenna, S. Low-valent germanium and tin hydrides as catalysts for hydroboration, hydrodeoxygenation, and hydrodesulfurization of heterocumulenes. *Dalton Trans.* **2024**, 53, 18207–18216.
15. Sarkar, N.; Sahoo, R. K.; **Mukhopadhyay, S.**; Nembenna, S., Organoaluminum Cation Catalyzed Selective Hydrosilylation of Carbonyls, Alkenes, and Alkynes. *Eur. J. Inorg. Chem.* **2022**, 2022 (8), e202101030.

Review Article & Book Chapters

16. Prajapati, D.J., Das, A., **Mukhopadhyay, S.** and Nembenna, S. Zinc-Catalyzed Hydrofunctionalization Reactions. *Encyclopedia of Inorganic and Bioinorganic Chemistry*; Wiley, **2026**. 10.1002/9781119951438.eibc2907
17. Nembenna, S.; **Mukhopadhyay, S.**; Rajput, S. Guanidinate ligand-stabilized alkaline earth metal complexes: Synthesis, reactivity, and catalytic applications. *Encyclopedia of Inorganic and Bioinorganic Chemistry*; Wiley, **2023**. 10.1002/9781119951438.eibc2852.
18. **Mukhopadhyay, S.**; Patro, A. G.; Vadavi, R. S.; Nembenna, S. Coordination chemistry of main-group metals with organic isocyanides. *Eur. J. Inorg. Chem.* **2022**, DOI: 10.1002/ejic.202200469
19. Nembenna, S.; Sarkar, N.; Sahoo, R. K.; **Mukhopadhyay, S.** Organometallic complexes of alkaline earth metals. *Comprehensive Organometallic Chemistry IV*; Elsevier, **2022**.